

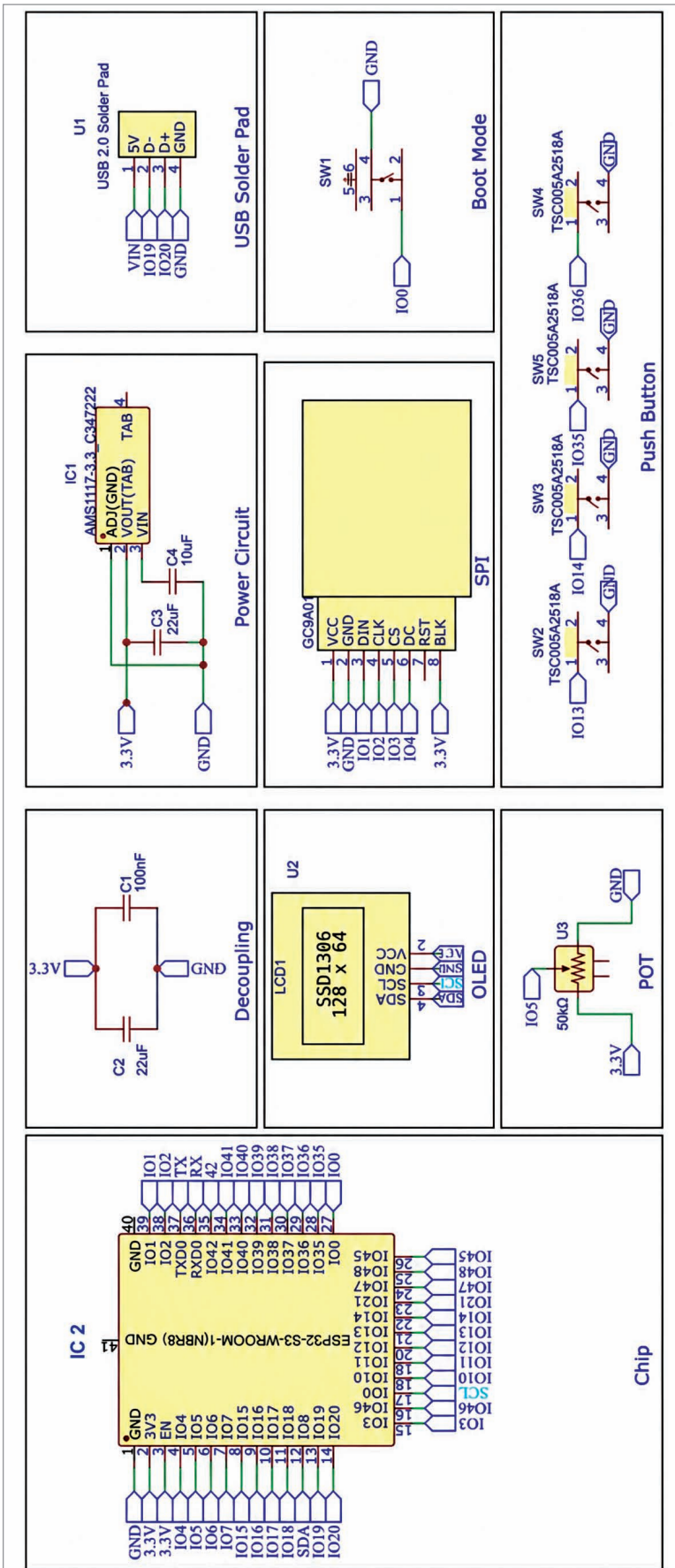


Fig. 2: Circuit diagram

controller, handling Wi-Fi connectivity, CSI processing, and display and user-interface control.

The regulator generates a stable 3.3V supply for the ESP32-S3 and other circuit sections. Capacitors C1 through C4 provide supply filtering and decoupling and are placed close to the regulator and ESP32-S3 for stable operation. The ESP32-S3 and display together consume less than 800mA when the ESP32-S3 operates in Wi-Fi AP or STA mode, leaving approximately 200mA of current margin.

The USB-C connector also provides D- and D+ connections to IO19 and IO20 of the ESP32-S3, respectively, for programming and debugging. Alternatively, the ESP32-S3 can be programmed off-board using a dedicated programming jig before mounting it on the PCB. A boot-control switch (SW1) connects IO0 to ground for programming-mode selection.

The PCB supports two display options. An SSD1306 128 × 64 OLED (U2) can be connected through the I²C interface using SDA and SCL, while a GC9A01 colour TFT display uses the SPI interface through DIN, CLK, CS, DC, RST, and BLK signals. The 50kΩ potentiometer is connected to IO5 and provides an analogue input for user control. Four tactile switches, SW2 through SW5, are connected to IO13, IO14, IO35, and IO36, respectively, and provide menu selection and navigation.

During operation, the ESP32-S3 receives Wi-Fi signals, acquires and processes CSI data, and displays the relevant signal, sensing, or indoor-location information on the selected display. All signal and power connections are routed through the PCB to form a compact standalone Wi-Fi sensing and analysis tool.

PCB design

The PCB is designed as a compact circular board, with the display and user-interface controls placed on the top side for easy access. The remaining components, including the ESP32-S3 module and 50kΩ potentiometer, are